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Skill Recommendation System and Resume Analysis using AI

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***Abstract*— The goal of the Skill Recommendation System and Resume Analysis is to improve resume writing skills and raise the probability of getting hired for a job. It offers specific guidance on how to make your CV stand out from the thousands of other applicants on internet job sites. The system generates tailored recommendations for course selection and skill improvement based on information it has extracted from resumes, including experience and talents. The suggested method aims to help job seekers improve their resumes and speed up the hiring process. The technology extracts relevant data from resumes using natural language processing (NLP) techniques and offers specific recommendations for course selection and improvement of skills. The aim of the system is to help companies find qualified applicants more quickly and help job seekers improve their resumes by automating the resume analysis process.**

Keywords*— *Intelligence, Natural language processing, Recommendation for skills, Data visualization.

I. INTRODUCTION

This paper analyzes AI methods for automating HR tasks in the hiring process, with a particular emphasis on resume parsing and candidate shortlisting according to a variety of criteria like work history, interests, and skill sets.[1] The technology expedites the decision-making process for reviewing resumes that have been shortlisted by using data visualization approaches to communicate selection outcomes. The objective of this system is to streamline the hiring process by automating the resume review process.

This project is an innovative application that automates the resume analysis process using techniques from artificial intelligence (AI) and natural language processing (NLP). This Streamlit-based online program allows users to upload their resumes, from which it parses them to extract crucial information such as contact details, skills, and experience.[2]

Moving from paper resumes to digital resumes has become common in the rapidly evolving corporate recruitment environment. Unfortunately, the present manual resume filling method takes a lot of time and frequently results in candidates not being able to get a job since their talents and the specifications of the position don't match. An intelligent recruitment system that makes use of AI and machine learning can match candidates to job needs and corporate culture by analyzing their resumes, abilities, and interests. By automating resume screening and shortlisting, not only increases job satisfaction and retention rates, but also optimizes the recruitment process.

An overview of the increasing importance of resume analysis in the hiring process is given in the introduction. It draws attention to the difficulties recruiters and job searchers face when manually assessing resumes and the possible advantages of employing AI to automate this process.[3]

II. LITERATURE SURVEY

In "Smart Resume Analyzer: A Case Study using RNN-based Keyword Extraction," the aim is to show how Natural Language Processing (NLP) can be used in a Smart Resume Analyzer to extract data from resumes, detect abilities, suggest job titles, offer customized resume improvement solutions, raise the likelihood that job seekers will find employment, provide ratings, make suggestions for changes, and store data in Python. The system's customized feedback is intended to revolutionize the application process and boost job seekers' productivity. [4]

The Natural Language Processing (NLP) perspective on resume analysis within the ambit of recruitment e-platforms sought to leverage the machine-learning approach to tacking the issue of manuals of resumes. Identifying candidates based on the analysis of the aforementioned profiles against keywords cued by relevant skill sets and deriding text from resumes is a hallmark description of natural language processing. [5]

The article "A Machine Learning approach for automation of Resume Recommendation System" proposes a framework in which machine learning techniques suggest resumes automatically. After categorizing resumes as per experience and skills, the system matches resumes on content basis and identifies the ones closely fitting into a particular job description. In this way, shortlisting candidates becomes all the less time-consuming and laborious for a recruiter.[6]

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The purpose of Modern Resume Analyzer For Students And Organizations is to help individuals design resumes that present their capabilities in a manner that augments their chances of securing employment. It leverages keyword matching algorithms to scan resumes, extract relevant information, and create a ranked list for interview invitations. By pinpointing recurring errors and offering advice on formatting, the system boosts the quality of the resume and increases the chances of interview invitations. [7]

III. PROPOSED SYSTEM

Three major steps of AI-based resume analysis is discussed in this methodology which revolves around the importance of feature extraction, data preprocessing, model training, and iterative validation to build a robust and adaptive resume analysis system. Then rephrase without copying from the reference [Refer Table. 1]

A. Data Gathering and Preparation

- Start building the resume data set comprised of huge diversity in content, format, and style.
- Preprocess resumes to extract relevant data such as contact information, education credentials, work experiences, and skills.
- Convert textual data to a structure that can be used for analysis.

B. Feature extraction:

- Relevant features for analysis are extracted from preprocessed resumes. This feature could be skill keywords, experience level, industry type, or job titles.
- Natural language processing techniques are employed to extract meaningful knowledge from the textual content of resumes.

C. Labeling data:

- Order the resumes according to quality, job skill suitability, or hiring qualification.
- Use labeled data to train the AI model to identify trends and predictions with respect to resume features.

D. Train the AI Model:

- Choose a relevant AI model, such as a decision tree or a neural network, to assess the resumes.
- Split the labeled datasets into sets for training and testing to assess the model's efficiency.
- Train the AI model using the training set while fine-tuning the parameters for maximum accuracy and functionality.

E. Integration and Deployment:

- Deploy the trained AI model live, such that it can be used to automate resume analysis.
- Database merging into recruitment procedures to bolster human decision-making and improve process efficiency.

F. New Resume Adaption Ability:

- Develop a system that is able to be adapted to new resumes with regular incorporating of new information into the AI model.
- Constantly make sure that the underlying model precisely reflects and is compatible with the structural and content change of resumes over time.[8]

TABLE I. PROPOSED METHODOLOGY FOR RESUME ANALYSIS

Serial No	Step	Description
1	Upload Resume	Provide a file upload interface for users to upload their resumes
2	Extract Text	Utilize PDF parsing libraries (e.g., PyMuPDF, pdfplumber) to extract text from the uploaded resume file.
3	Parse Resume	Use a resume parsing library (e.g., Pyresparser) to extract relevant information (e.g., skills, experience, education) from the extracted text.
4	Analyze Skills	Apply natural language techniques to carry out analysis of the skills being extracted, and categorize the skills under the appropriate category.
5	Display Results	Apply natural language processing (NLP) methods to examine the extracted skills, categorize them to suitable categories.
6	Recommend Courses	Based on the skills analysis, recommend relevant courses or certifications to enhance the user's skillset.
7	Store Data	Accordingly, recommend basic courses or certifications to improve the user's skills based on skill analytics. [9]

IV. SYSTEM ARCHITECTURE

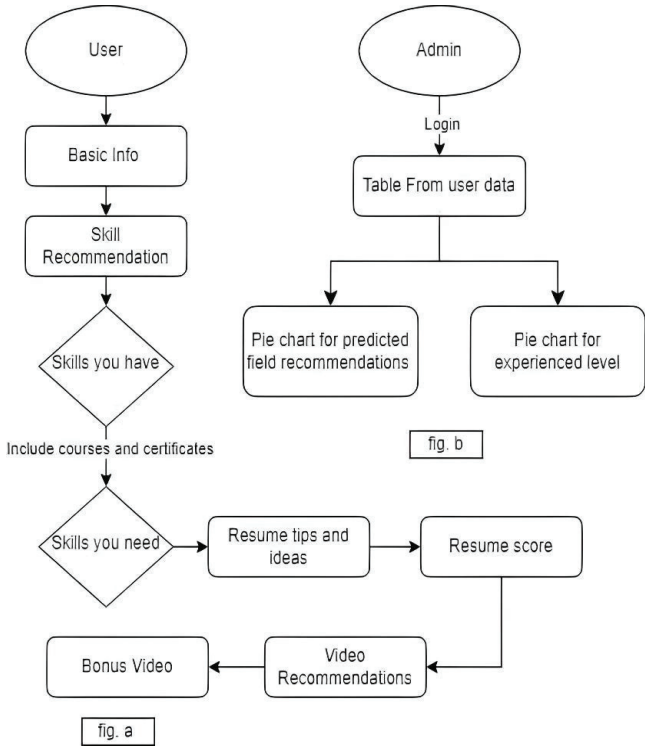


Fig. 1. Workflow of Skill Recommendation System and Resume Analysis.

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V. RESULT AND DISCUSSION

The Resume Analysis and Skill Recommendation System have shown positive results, offering accurate analysis and suggestions to improve users' resumes. The abilities Recommendation part of the system recommends particular abilities that can greatly increase a resume's chances of getting hired. The system also provides customers with Resume Tips & Ideas, which gives them pointers on how to improve their resumes even more.[10]

Users can get a score from the Resume Score feature that represents the general efficacy and quality of their resumes. In addition, the system offers a Bonus Video for Resume Writing Tips, giving customers access to extra tools to enhance their resumes.

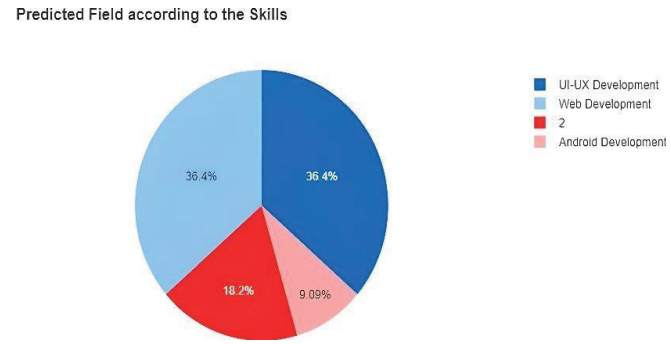


Fig. 2. Results for Predicted Field Recommendations

Based on the skills taken from the resume analysis, the expected fields are shown. Refer [fig. 2] for predicted Field Recommendations.[11] Such as the chart would show greater percentages for the domains of web development and UI/UX design if a significant number of resumes mentioned these talents. Managers can more clearly understand the trends in hiring and adjust their approaches to candidate selection and job listings by using this chart.

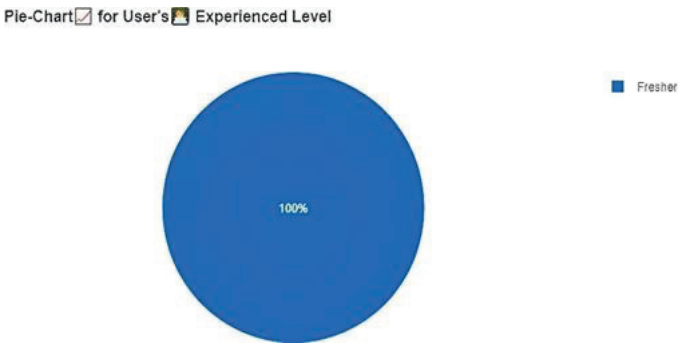


Fig. 3. Results for User's Experienced Level

The proposed level of experience of the candidates is shown in this User's Experienced level Refer [Fig. 3], which classifies them as either freshers or experienced workers or students.[12] As a result, the administrator can better understand the backgrounds of job seekers and make decisions regarding job listings and resume guidance.

For each resume, the Smart Resume Analyzer provides a complete analysis and provides a full description of the candidate's qualifications.[13]

	ID	Name	Email	Resume Score	Timestamp
0	12	Rohit Nandagawali	nandagawalirohit143@gmail.com	40	2024-03-21_19:5
1	13	ANUJ PUJARI	anujbp.2121@gmail.com	60	2024-03-21_19:5
2	14	Prajwal Tribhuvan	prajwal.22010464@viit.ac.in	20	2024-03-21_20:0
3	15	Bharat Shinde	bharat.22220076@viit.ac.in	40	2024-03-21_20:0
4	19	Sanket Ramchandra Satpute	202201918@vupune.ac.in	20	2024-03-21_20:1

Fig. 4. Resume Report

VI. CONCLUSION

Job seekers, especially recent graduates starting their careers, can benefit greatly from the Skill Recommendation System and Resume Analysis. Through the provision of particular recommendations for which abilities to highlight on resumes and advice on how to make resumes look better overall, these systems increase your chances of getting hired. Also, for those looking to differentiate their resume from the competition, the systems' capacity to offer insights into job market trends and resume competition can be extremely helpful. All things considered, these systems provide a useful and successful means of optimizing resumes and raising one's chances of obtaining a job.

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